

3. (a) The table shows information about some organic compounds.

Name	Molecular formula	Structure	Homologous series
ethanol	C_2H_5OH		alcohols
ethanoic acid	CH_3COOH	<pre> H H — C — C // \ H O O — H </pre>	
.....	C_3H_6	<pre> H H H H — C = C — C — H H H H </pre>	alkenes

(i) **Complete the table.**

[3]

(ii) The molecular mass of an alkene is 98. Give the molecular formula of this alkene.

[1]

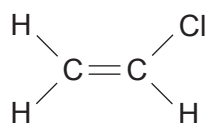
$$A_r(C) = 12$$

$$A_r(H) = 1$$

Molecular formula

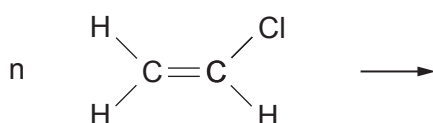


- (b) Polyvinyl chloride (PVC) is formed from the following monomer in a polymerisation reaction.



Complete the equation for the polymerisation reaction.

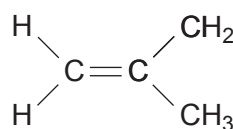
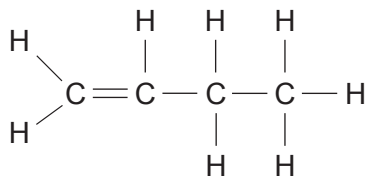
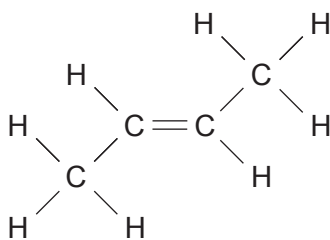
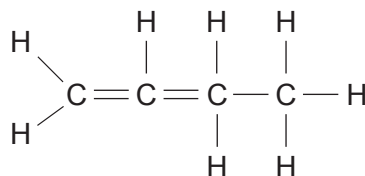
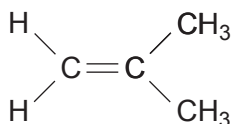
[2]



- (c) A student was asked to draw structures for isomers with the molecular formula C_4H_8 .

She drew the following diagrams. Circle **all** the **correct** structures.

[2]



(d) Pent-2-ene reacts with bromine water in a similar manner to ethene.

Complete the equation for the reaction of pent-2-ene with bromine by drawing the structure of the product.

[1]

